ASTRA HEALTH: An AI-Powered Astronaut Health Monitoring and Decision-Support Platform

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Abstract

Long-duration human spaceflight exposes astronauts to space radiation, isolation and confinement, altered gravity, distance from Earth, and hostile closed environments. NASA's Human Research Program identifies these five hazards and notes that they can compound one another. As missions travel farther from Earth, medical support becomes harder to receive, so crews and medical teams need health information that is organized, personalized and easy to interpret.

This report presents ASTRA HEALTH, an AI-powered health monitoring and decision-support platform designed to meet that need. The platform combines a monitoring dashboard, personalized health baselines, AI-based analysis, an AI assistant, preventive recommendations, exercise guidance, a mission planner, and a structured registry of medical reference standards. Its central design idea is that user-specific health data and medical knowledge are joined in one system, so readings are interpreted in context rather than displayed in isolation.

The report describes the underlying problem using NASA sources, the system design, the role of reference standards such as NASA-STD-3001, the expected impact, and the alignment with the NASA Space Apps Challenge 2026 "Health Monitoring Software" challenge. ASTRA HEALTH is a decision-support tool. It does not replace professional medical judgment, and no clinical validation is claimed.

Keywords: astronaut health, health monitoring, decision support, artificial intelligence, spaceflight hazards, reference standards, preventive care

1. Introduction

NASA's plans for exploration include a return to the Moon and journeys to Mars. Keeping crews healthy on such missions depends on understanding how the human body and mind respond to the spaceflight environment. NASA's Human Research Program (HRP)

studies this through ground-based analogs, laboratories and the International Space Station, and has organized the main challenges into five hazards [1][5].

One consequence of distance is that real-time support from Earth declines. Astronauts and the teams who care for them increasingly need to make sense of health data themselves. Raw readings do not meet that need: a number is useful only when it is compared with the right reference, followed over time, and explained clearly.

ASTRA HEALTH is a software platform built around this idea. Instead of showing health readings alone, it aims to turn them into meaningful insight that helps astronauts and medical teams understand potential health changes and take appropriate preventive action.

This report makes four contributions:

1. It summarizes the spaceflight hazards and health risks that motivate continuous monitoring, using NASA sources.
2. It describes an integrated design that joins monitoring, personalized baselines, AI analysis, an AI assistant and a medical standards registry.
3. It explains how reference standards such as NASA-STD-3001 can ground the platform's interpretations.
4. It sets out the expected impact, the alignment with the NASA Space Apps Challenge 2026, and the limits of the work.

2. Background: spaceflight hazards and health risks

NASA's HRP groups the challenges of human spaceflight into five hazards: space radiation, isolation and confinement, distance from Earth, gravity fields, and hostile or closed environments. The hazards do not stand alone. They can feed off one another and worsen effects on the human body [1]. A NASA-authored review reports that these hazards are linked with over 30 human health risks documented by the HRP [5].

Hazard	Description from NASA	Implication for monitoring
Space radiation	Invisible, and considered one of the most hazardous aspects of spaceflight [1]	Long-term risk builds quietly, so exposure-related context should be tracked over time

Hazard	Description from NASA	Implication for monitoring
Isolation and confinement	Behavioral responses among groups isolated in a small space, far from Earth, over a long period [1]	Changes in sleep, mood and routine need early attention
Distance from Earth	Instructions, supplies and medical care become harder to receive from Earth [1]	Crews must interpret their own health data with less real-time support
Gravity fields	Muscles, bones, inner ear and organs must adjust to new gravity, and again on return [1]	Cardiovascular, muscle and bone changes need consistent tracking
Hostile and closed environments	Temperature, pressure, lighting, noise, microbes and immune responses shape the habitat [1]	Environmental and physiological signals interact and are hard to follow separately

NASA gives the distance hazard a concrete scale: Mars is about 140 million miles from Earth on average, and a Mars mission would last roughly three years, which makes planning and self-sufficiency essential [1]. NASA's risk pages also name specific human-system risks, including cardiovascular adaptations, spaceflight-associated neuro-ocular syndrome, spaceflight-induced bone changes, and inadequate sleep with irregular schedules [2].

NASA has also found that nutrition and exercise help protect crew members from bone, muscle and cardiovascular deconditioning [3]. This finding motivates the exercise-management component of the platform described in Section 4.

3. Problem statement and objectives

Traditional approaches to health monitoring may not provide a unified, personalized and continuously updated understanding of an individual astronaut's health. Measurements can be scattered, ranges built for people on Earth may not suit astronauts, and interpreting the data takes time and expertise that may not be at hand during a long mission.

The problem addressed here can be stated as follows: how can health information be organized, unusual changes identified, readings interpreted against appropriate references, and preventive management supported, all in one system that astronauts and medical teams can use?

From this statement, ASTRA HEALTH pursues five objectives:

1. Gather and organize health indicators in one place and follow them over time.
2. Identify potentially unusual changes against each user's own baseline.
3. Interpret readings with medical reference standards, separating general human ranges from astronaut-specific considerations.
4. Explain results in context through an AI assistant that uses the user's own data.
5. Turn insight into action through preventive recommendations, exercise guidance and activity planning.

4. System design and methodology

4.1 Design approach

ASTRA HEALTH is designed as one integrated platform. User-specific health information and medical knowledge meet in a shared structure, so the analysis, the AI assistant and the recommendations all rely on the same data and the same reference standards. The path from data to decision has five steps:

1. **Monitor.** Health indicators and physiological measurements are collected and shown on a dashboard.
2. **Personalize.** Each user's own baseline becomes the reference point for what counts as a change.
3. **Analyze.** AI-based analysis highlights unusual changes and potential anomalies, and reads them against medical reference standards.
4. **Explain.** The AI assistant uses the user's data and the same standards to explain what readings may mean.
5. **Act.** Preventive recommendations, exercise guidance and a mission planner turn insight into routines.

4.2 System components

Component	Function	Objective served
Health monitoring dashboard	Shows measurements, charts and trends over time, and gives an overview of health status	1
AI-based health analysis	Identifies unusual changes and potential anomalies and interprets readings against reference standards	2, 3
AI assistant	Lets users discuss their health information and explains recorded measurements using their own data and the standards	4
Personalized recommendations	Offers preventive suggestions and shows which areas may need attention	5
Exercise and physical health management	Gives exercise information, explains expected benefits and the consequences of consistently missed sessions, and supports awareness of cardiovascular, muscle and bone health	5
Mission planner and activity tracking	Organizes missions and activities and tracks health-related routines	5
Medical knowledge and reference standards	Holds the standards used for interpretation, separates general and astronaut-specific considerations, and keeps thresholds consistent	3

4.3 Technical approach

The implementation combines software development, backend data management, AI integration and medical knowledge. Its elements are a web-based interface with an interactive dashboard, user authentication with personalized profiles, backend APIs with database integration, AI-powered analysis and conversational assistance, a structured

medical standards and knowledge registry, data visualization and tracking, and personalized recommendations based on the information available for each user. This report does not name specific technologies, algorithms, sensors or integrations, because none has been confirmed for the project.

5. Medical knowledge and reference standards

A reading means little without a reference. Reference ranges that suit people on Earth may not suit astronauts, so ASTRA HEALTH keeps its medical knowledge in a structured registry that separates general human reference ranges from astronaut-specific considerations. The same registry feeds the analysis and the AI assistant, so thresholds and interpretations stay consistent.

NASA-STD-3001, Volume 1 (Crew Health), is the most relevant official source for this kind of registry. Its current revision, Revision C, was approved by NASA's Office of the Chief Health and Medical Officer on 15 September 2023 [4]. The standard sets requirements for fitness for duty, space permissible exposure limits, permissible outcome limits, levels of medical care, medical diagnosis, intervention, treatment and care, and countermeasures. It applies to the pre-mission, in-mission and post-mission phases of human spaceflight, and it treats human physiologic parameters as a system [4]. It is also written to complement current medical standards of clinical practice, which supports a registry that distinguishes astronaut-specific limits from general medical practice [4].

In practice, each threshold in the registry should cite its source so that users and medical teams can see where an interpretation comes from. Reference standards in the registry are intended to be reviewed by qualified medical experts before any operational use.

6. Expected impact and discussion

ASTRA HEALTH aims to contribute to safer and more informed human spaceflight. The benefits below are expected, not measured: the project has not been clinically validated, and no results are claimed.

- **Organized, continuous monitoring.** Health information is gathered in one place and followed over time instead of being scattered across separate readings.
- **Earlier attention to unusual changes.** Comparing readings with a personal baseline and with reference standards can help surface potentially unusual changes sooner.
- **Better understanding.** The dashboard and the AI assistant help astronauts understand what their physiological information may mean.

- **Preventive action.** Recommendations and exercise guidance connect insight to daily routines, including the cardiovascular, muscle and bone health that NASA associates with exercise [3].
- **Accessible medical knowledge.** Reference standards are available where a reading is viewed, which matters when medical support from Earth is delayed or limited [1].
- **Support for decisions.** Astronauts and medical teams gain a clearer basis for health decisions during missions, with professional judgment kept in the loop.

The main strength of the design is integration. Because analysis, assistant and recommendations share one data set and one standards registry, a reading is flagged, explained and acted on consistently. A dashboard alone would show numbers, and a stand-alone chatbot would lack the user's own data and a stable set of references. The main open questions concern validation, which Section 8 discusses.

7. Alignment with the NASA Space Apps Challenge 2026

The 2026 challenge, "Create Health Monitoring Software for Astronauts on Space Missions," asks teams to build health monitoring software that gathers health indicators and enables astronauts to evaluate and act on the status of their health [6]. ASTRA HEALTH answers each part of that request with a specific capability.

Challenge element	How ASTRA HEALTH addresses it
Gather health indicators and monitor astronaut health	The dashboard collects measurements in one place and tracks them over time against a personal baseline
Identify potential health changes through data analysis	AI-based analysis flags unusual changes and potential anomalies in the available data
Interpret data and give personalized insight	The AI assistant explains readings using the user's own data, so answers are specific to that person
Ground interpretation in medical knowledge	A standards registry separates general human ranges from astronaut-specific considerations and feeds both analysis and assistant

Challenge element	How ASTRA HEALTH addresses it
Let astronauts act on their health status	Preventive recommendations, exercise guidance and the mission planner turn insight into routines and decisions

The approach is relevant because the hazards NASA describes affect many body systems at once and can compound [1], and because crews far from Earth have less real-time medical support. A platform that keeps data, standards and guidance together serves that situation better than separate readings or a stand-alone chatbot.

8. Limitations and responsible use

ASTRA HEALTH is a decision-support tool. It is not a medical device, it does not diagnose or treat, and it does not replace professional medical judgment.

- **No clinical validation.** The project makes no claim of clinical validation, improved outcomes or prevented emergencies.
- **Data limits.** Analysis can only be as good as the health information available to it. Any demonstration relies on the data the team actually used.
- **AI limits.** AI-generated interpretations can be wrong. Outputs should be presented as aids to understanding, with the reference standard behind each one visible to the user.
- **Reference ranges.** Thresholds must be traceable to a named source and reviewed by qualified medical experts before any real use.
- **Scope of this report.** It describes the platform's designed capabilities and does not report performance results or clinical outcomes.

9. Future work

The following steps are proposed directions, not completed work.

1. **Expert review of the standards registry.** Have space-medicine specialists check each threshold and its source.
2. **Testing with public spaceflight data.** Run the analysis on openly published spaceflight datasets, such as those in NASA's Open Science Data Repository, and report the results descriptively.

3. **Closer fit to NASA-STD-3001.** Map registry entries to specific requirements and limits in Volume 1, and keep the mapping current as new revisions are issued [4].
4. **Device data import.** Study ways to bring in readings from wearable or medical devices, once suitable sources are identified.
5. **Usability testing.** Gather feedback from target users on the clarity of the dashboard, recommendations and assistant answers.
6. **Formal validation.** If the project moves beyond a prototype, plan validation appropriate to a decision-support tool before any operational use.

10. Conclusion

Spaceflight hazards affect many body systems at once, and crews far from Earth have less real-time medical support [1]. ASTRA HEALTH responds with an integrated platform that gathers health indicators, compares them with personal baselines and medical reference standards, flags potentially unusual changes, explains readings through an AI assistant, and supports preventive action through recommendations, exercise guidance and activity planning.

Its value lies in the integration of these parts around shared data and shared standards. The platform is a decision-support tool, not a clinical one, and its expected benefits are realistic aims rather than proven results. With expert review of its reference standards and appropriate validation, it could become a useful aid to astronaut health awareness and to health-related decisions on future missions.

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